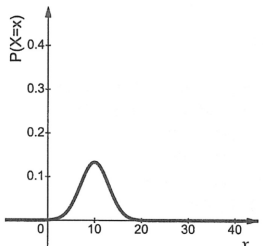
# The Normal Distribution

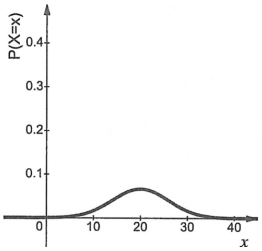
Questions 1–65

**Question 1.** The four diagrams below show $X \sim \mathrm{N}(10, 1)$, $X \sim \mathrm{N}(10, 3)$, $X \sim \mathrm{N}(20, 3)$ and $X \sim \mathrm{N}(20, 6)$. Which is which?

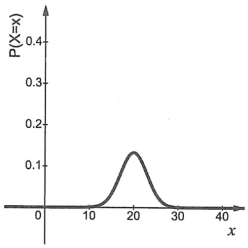
A



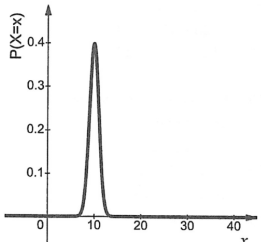

B



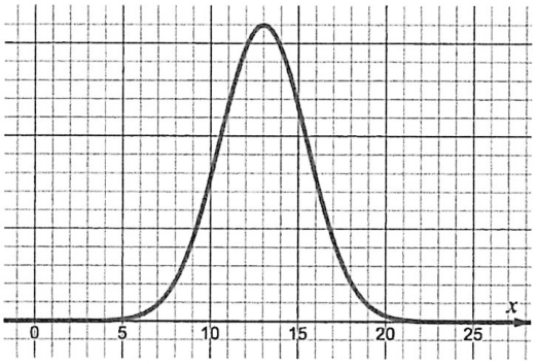

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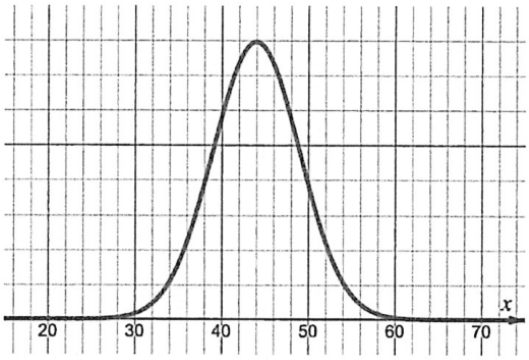

D




**Question 2.** It is given that the distribution shown below is normal. Use the diagram to estimate the standard deviation.



**Question 3.** It is given that the distribution shown below is normal. Use the diagram to estimate the standard deviation.




**Question 4.** Find the standard deviation of each of these normal distributions:

a) $X \sim \mathrm{N}(72, 49)$
b) $X \sim \mathrm{N}(12, 5)$
c) $X \sim \mathrm{N}(53, \sqrt{2})$

**Question 5.** Approximately 99.73% of the area under a normal distribution is within three standard deviations of the mean. For each of these normal distributions find the values of $a$ and $b$ such that $\mathrm{P}(a < X < b) \approx 0.9973$:

a) $X \sim \mathrm{N}(30, 25)$
b) $X \sim \mathrm{N}(18, 12.25)$
c) $X \sim \mathrm{N}(605, 1024)$

**Question 6.** Sketch the following normal distributions. You do not need to consider their heights.

a) $X \sim \mathrm{N}(25, 5^2)$
b) $X \sim \mathrm{N}(100, 100)$
c) $X \sim \mathrm{N}(50, 5)$

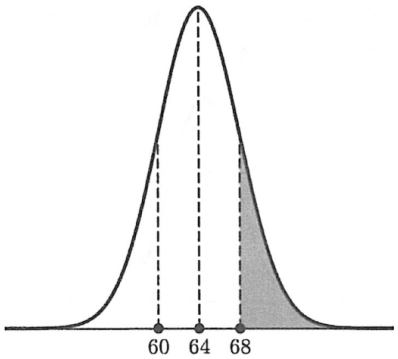
**Question 7.** Sketch $X \sim \mathrm{N}(60, 5^2)$ and $Y \sim \mathrm{N}(100, 5^2)$ on the same axes, identifying which is which.

**Question 8.** Sketch $X \sim \mathrm{N}(20, 5^2)$ and $Y \sim \mathrm{N}(30, 5^2)$ on the same axes, identifying which is which.

**Question 9.** Sketch $X \sim \mathrm{N}(60, 10^2)$ and $Y \sim \mathrm{N}(60, 5^2)$ on the same axes, identifying which is which.

**Question 10.** Sketch $X \sim \mathrm{N}(8, 1)$ and $Y \sim \mathrm{N}(10, 4)$ on the same axes, identifying which is which.

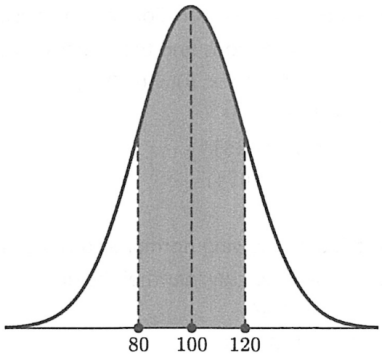
**Question 11.** It is given that $X \sim \mathrm{N}(64, 4^2)$. The graph of $X$ is shown below and it is given that $\mathrm{P}(X > 68) = 0.158655$




Use the symmetry of the graph to find:

a) $\mathrm{P}(X < 68)$
b) $\mathrm{P}(X < 60)$
c) $\mathrm{P}(60 < X < 68)$

**Question 12.** It is given that $X \sim \mathrm{N}(100, 400)$. The graph of $X$ is shown below and it is given that $\mathrm{P}(80 < X < 120) = 0.682690$




Use the symmetry of the graph to find:

a) $\mathrm{P}(X < 80)$
b) $\mathrm{P}(X > 120)$
c) $\mathrm{P}(80 < X < 100)$

**Question 13.** It is given that $X \sim \mathrm{N}(8, 4)$ and $Z \sim \mathrm{N}(0, 1)$. Find the corresponding values of $Z$ when:

a) $X = 7$

b) $X = 9$

c) $X = 10.2$

**Question 14.** It is given that $X \sim \mathrm{N}(16, 16)$. By converting to the standard normal distribution, find:

a) $\mathrm{P}(X < 15)$

b) $\mathrm{P}(X > 17)$

c) $\mathrm{P}(13 < X < 15)$

d) $\mathrm{P}(18 < X < 19)$

**Question 15.** Given $X \sim \mathrm{N}(40, 8^2)$, use a calculator to find:

a) $\mathrm{P}(X \leq 37)$
b) $\mathrm{P}(X < 37)$
c) $\mathrm{P}(X \geq 45)$
d) $\mathrm{P}(X > 45)$

**Question 16.** Given $X \sim \mathrm{N}(800, 45^2)$, use a calculator to find:

a) $\mathrm{P}(X < 860)$
b) $\mathrm{P}(X > 810)$
c) $\mathrm{P}(780 < X < 850)$
d) $\mathrm{P}(860 < X < 880)$

**Question 17.** Given $X \sim \mathrm{N}(45, 3^2)$, find:

a) $\mathrm{P}(X = 46)$

b) $\mathrm{P}(X \neq 40)$

**Question 18.** Given $X \sim \mathrm{N}(80, 30)$, find:

a) $\mathrm{P}(X < 80)$

b) $\mathrm{P}(X < 75)$

c) $\mathrm{P}(X > 83)$

d) $\mathrm{P}(75 < X < 83)$

**Question 19.** Given $X \sim \mathrm{N}(125, 88)$, find:

a) $\mathrm{P}(X < 110)$

b) $\mathrm{P}(125 < X < 130)$

c) $\mathrm{P}(130 < X < 135)$

**Question 20.** Use the standard normal distribution, $Z \sim \mathrm{N}(0, 1)$ to find:

a) $\mathrm{P}(-1 < Z < 1)$

b) $\mathrm{P}(-2 < Z < 2)$

c) $\mathrm{P}(-3 < Z < 3)$

d) $\mathrm{P}(-4 < Z < 4)$

**Question 21.** Given $X \sim \mathrm{N}(200, 15^2)$ and your answers to the previous question, find:

a) $\mathrm{P}(185 < X < 215)$

b) $\mathrm{P}(170 < X < 230)$

c) $\mathrm{P}(155 < X < 245)$

d) $\mathrm{P}(140 < X < 260)$

**Question 22.** Given $X \sim \mathrm{N}(\mu, \frac{1}{4}\mu^2)$, find $\mathrm{P}(X < 0.9\mu)$

**Question 23.** Given $X \sim \mathrm{N}(5\mu, 16\mu^2)$, find $\mathrm{P}(X > \frac{11}{2}\mu)$

**Question 24.** Given $X \sim \mathrm{N}(2\mu, 81\mu^2)$, find $\mathrm{P}(0.1\mu < X < 0.5\mu)$

**Question 25.** 200 university students are surveyed and it is found that the monthly rent spent on student accommodation, £$X$, had the following results, where $\bar{x}$ denotes the sample mean:

$$\sum x = 105000 \text{ and } \sum (x - \bar{x})^2 = 440000$$

a) Find the sample mean and sample standard deviation of $X$
b) Using a normal distribution to model $X$, find the probability that a university student will pay more than £535 each month.

**Question 26.** 100 sixth form students are surveyed, and it is found that the number of hours of driving lessons, $X$, had the following results:

$$\sum x = 4650 \text{ and } \sum x^2 = 219000$$

a) Find the sample mean, $\bar{x}$, of $X$
b) It is given that the sample variance,

$$s^2 = \frac{\sum x^2}{n-1} - \frac{(\sum x)^2}{n(n-1)}$$

Find $s^2$
c) Assuming that $X$ be modelled by $\mathrm{N}(\bar{x}, s^2)$, find $\mathrm{P}(43 < X < 46)$

**Question 27.** The heights of male NBA basketball players have a mean of 2.01 metres and a standard deviation of 10 cm. It is known that 99% of the basketball players are between 1.79 m and 2.23 m. Comment on whether it would be suitable to model the heights of male NBA basketball players using a normal distribution.

**Question 28.** Using the standard normal distribution, $Z \sim \mathrm{N}(0, 1)$,

a) Find $a$ such that $\mathrm{P}(Z < a) = 0.25$
b) Find $b$ such that $\mathrm{P}(Z < b) = 0.9$
c) Find $c$ such that $\mathrm{P}(Z > c) = 0.01$
d) Find $d$ such that $\mathrm{P}(Z > d) = 0.8$

**Question 29.** Given $X \sim \mathrm{N}(27, 4^2)$,

a) Find $a$ such that $\mathrm{P}(X < a) = 0.02$

b) Find $b$ such that $\mathrm{P}(X < b) = 0.7$

c) Find $c$ such that $\mathrm{P}(X > c) = 0.21$

d) Find $d$ such that $\mathrm{P}(X > d) = 0.6$

**Question 30.** It is given that $X \sim \mathrm{N}(800, 900)$ and $\mathrm{P}(a < X < b) = 0.9$. It is also given that $\mathrm{P}(a < X < 800) = \mathrm{P}(800 < X < b)$. Find the value of $a$ and the value of $b$.

**Question 31.** It is given that $X \sim \mathrm{N}(65, 12)$ and $\mathrm{P}(64 < X < a) = 0.25$. Find the value of $a$

**Question 32.** It is given that $X \sim \mathrm{N}(18, 10)$.

a) Find $a$ given that $\mathrm{P}(X > a) = 0.3$

b) Find $b$ given that $\mathrm{P}(18 - b < X < 18 + b) = 0.3$

**Question 33.** It is given that $X \sim \mathrm{N}(\mu, 10^2)$. Find the value of $\mu$ in each of these cases:

a) $\mathrm{P}(X < 40) = 0.5$
b) $\mathrm{P}(X < 50) = 0.3$
c) $\mathrm{P}(X > 200) = 0.7$
d) $\mathrm{P}(X > 70) = 0.01$

**Question 34.** It is given that $X \sim \mathrm{N}(33, \sigma^2)$. Find the value of $\sigma$ in each of these cases:

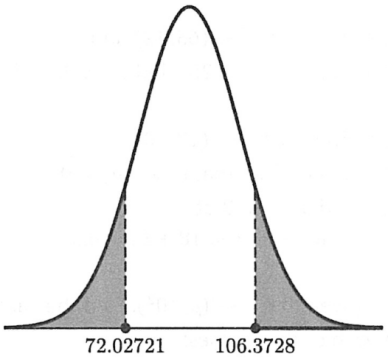
a) $\mathrm{P}(X < 30) = 0.2$
b) $\mathrm{P}(X < 50) = 0.6$
c) $\mathrm{P}(X > 1) = 0.99$
d) $\mathrm{P}(X > 34) = 0.49$

**Question 35.** The average monthly food bill, £$X$, for a family of 4 in the UK can be modelled by a normal distribution with mean £725. It is known that $\mathrm{P}(X < 700) = 0.35$. Find the standard deviation of $X$ to one decimal place.

**Question 36.** Loose leaf tea is packaged in sachets. The mass, $X$, of loose leaf tea in each sachet is known to be normally distributed with a mean of 3 grams. It is given that 50% of the masses are between 2.6 g and 3.4 g. Determine the standard deviation of the masses.

**Question 37.** Category 2 ambulance calls are classified as an emergency that may require immediate assessment or on-scene intervention. Response times should have an average of 18 minutes and 90% must be responded to within 40 minutes. Explain why response times may not be normally distributed.

**Question 38.** The distribution $X \sim \mathrm{N}(\mu, \sigma^2)$ is shown below. The two shaded regions add to an area of 0.2.




a) Find $\mu$ and $\sigma^2$, given that the lower boundary is 72.02721 and the upper boundary is 106.3728
b) It is given that $Y$ is the random variable such that $Y = 3X - 2$. Write down the distribution of $Y$.

**Question 39.** It is given that $X \sim \mathrm{N}(84, 5.29)$ can be transformed to $Y \sim \mathrm{N}(326.8, 72.4201)$ by the transformation $y = ax + b$. Find the values of $a$ and $b$.

**Question 40.** It is given that $X \sim \mathrm{N}(\mu, \sigma^2)$. It is also given that $\mathrm{P}(X < 99) = 0.5$ and $\mathrm{P}(X > 90) = 0.6$. Find the value of $\mu$ and the value of $\sigma$

**Question 41.** It is given that $X \sim \mathrm{N}(\mu, \sigma^2)$. It is also given that $\mathrm{P}(X > 11) = 0.5$ and $\mathrm{P}(X > 12) = 0.05$. Find the value of $\mu$ and the value of $\sigma$

**Question 42.** It is given that $X \sim \mathrm{N}(\mu, \sigma^2)$. Find the value $\mu$ and $\sigma$ in each of these cases:

a) $\mathrm{P}(X < 10) = 0.3$ and $\mathrm{P}(X < 11) = 0.35$
b) $\mathrm{P}(X < 50) = 0.4$ and $\mathrm{P}(X < 90) = 0.65$
c) $\mathrm{P}(X < 3) = 0.01$ and $\mathrm{P}(X > 9) = 0.02$
d) $\mathrm{P}(X > 67) = 0.44$ and $\mathrm{P}(X > 73) = 0.33$

**Question 43.** A banana is considered small if it weighs less than 140 grams, medium if it weighs between 140 and 215 grams, and large if it weighs over 215 grams. 14% of bananas are found to be small and 14% are found to be large. By writing down an appropriate distribution, estimate the weight of the lightest banana.

**Question 44. Prerequisite knowledge for Q44–Q45: Points of Inflection, The Chain Rule & The Product Rule** The normal distribution with mean 0 and standard deviation 1 has equation

$$y = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}x^2}$$

Use differentiation to find the $x$-coordinates of the points of inflection.

**Question 45.** The normal distribution with mean $\mu$ and standard deviation $\sigma$ has equation

$$y = \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Use differentiation to find the $x$-coordinates of the points of inflection.

**Question 46. Prerequisite knowledge: Conditional Probability** Given $X \sim \mathrm{N}(75, 10)$, find:

a) $\mathrm{P}(X < 70 \mid X < 80)$
b) $\mathrm{P}(X > 70 \mid X < 76)$
c) $\mathrm{P}(X < 77 \mid X > 73)$
d) $\mathrm{P}(X > 81 \mid X > 80)$

**Question 47.** A nursery grows yucca plants. When they are ready to be sold, their heights are normally distributed with a mean of 75 cm and standard deviation 5 cm. One yucca plant is selected at random. Find the probability that it is taller than 80 cm given that it is taller than 76 cm.

**Question 48.** The weight of sandbags is known to be normally distributed with a mean of 33 kg and standard deviation of 3.5 kg. A sandbag is selected at random. Given that it weighs more than 34 kg, find the probability that it weighs less than 36 kg.

**Question 49. Prerequisite knowledge for Q49: The Binomial Distribution** The weight, $X$, of a farm's eggs may be modelled by a normal distribution with mean 49 grams and variance 9 grams.

a) A single egg is randomly selected. Find the probability that the egg has a weight of less than 47 grams.

b) Eggs are put in boxes of twelve before they are transported to the local supermarket. A box of twelve eggs is randomly selected. Find the probability that at least four of the eggs has a weight of less than 47 grams.

**Question 50.** The lifetime, $X$, of a discount battery may be modelled by a normal distribution with mean 10 hours and standard deviation 1.2 hours.

a) A single discount battery is randomly selected. Find the probability that the battery has a lifetime:

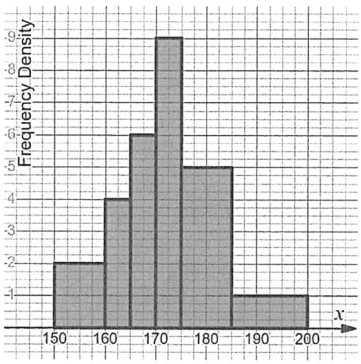
   i) equal to 11 hours;

   ii) greater than 11 hours.

b) Discount batteries are sold in packs of eight. A pack is randomly selected. Find the probability that at least two of the batteries have a lifetime of more than 11 hours.

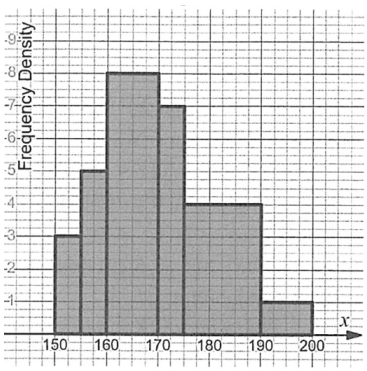
**Question 51.** The width, $W$, of a type of potato is known to be normally distributed with a mean of 5.7 cm and standard deviation 0.9 cm. A farmer is able to sell their potatoes to a large supermarket if $4.8 \leq W \leq 6.2$ for £0.25 per kilogram. If $W < 4.8$, the farmer sells the potatoes to a local farmer for £0.11 per kilogram. If $W > 6.2$, the farmer sells the potatoes to a local greengrocer for £0.21 per kilogram. Calculate how much the farmer will be able to sell a crop of 150 kg of potatoes for.

**Question 52.** The heights of 180 students were measured to the nearest cm and recorded in the histogram below.




a) Find $\mathrm{P}(x < 170)$
b) Find estimates for the mean and sample standard deviation of the heights.
c) Using your answers to b), model the data as a normal distribution, $Y \sim \mathrm{N}(\mu, \sigma^2)$. Find $\mathrm{P}(Y < 170)$ and thence comment on whether the normal distribution is an appropriate model for the data.

**Question 53.** The heights of a different group of 225 students were measured to the nearest cm and recorded in the histogram below.




a) Find $\mathrm{P}(x < 170)$
b) Find estimates for the mean and sample standard deviation of the heights.
c) Using your answers to b), model the data as a normal distribution, $Y \sim \mathrm{N}(\mu, \sigma^2)$. Find $\mathrm{P}(Y < 170)$ and thence comment on whether the normal distribution is an appropriate model for the data.

**Question 54.** Describe the conditions for which the binomial distribution $X \sim \mathrm{B}(n, p)$ can be approximated using the normal distribution $Y \sim \mathrm{N}(\mu, \sigma^2)$

**Question 55.** For each of the following binomial distributions, write down a normal distribution that could be used to approximate it, and comment on whether this would be appropriate.

a) $X \sim \mathrm{B}(500, 0.48)$
b) $X \sim \mathrm{B}(4, 0.5)$
c) $X \sim \mathrm{B}(150000, 0.001)$

**Question 56.** A circular spinner has five sections labelled 1, 2, 3, 4 and 5. The section labelled 2 is a sector with an angle of 160°. The spinner is spun 2000 times and the number, $X$, of times 2 is chosen by the spinner is recorded.

a) Write down the distribution of $X$ and explain why it would be appropriate to approximate the distribution of $X$ using a normal distribution.
b) Using a normal distribution to approximate the distribution of $X$, find $a$ such that $\mathrm{P}(X > a) \approx 0.2$
c) Using the distribution of $X$ and your value of $a$, find the actual value of $\mathrm{P}(X > a)$ to 3sf.

**Question 57.** It is given that $X \sim \mathrm{N}(300, 20^2)$ and $Y \sim \mathrm{N}(400, 30^2)$ where

$$\mathrm{P}(X > a) = \mathrm{P}(Y < a) = b$$

Find the value of $a$ and the value of $b$

**Question 58.** It is given that $X \sim \mathrm{N}(16.5, 5.29)$ and $Y \sim \mathrm{N}(20.2, 6.76)$ where

$$\mathrm{P}(X > a) = \mathrm{P}(Y < a) = b$$

Find the value of $a$ and the value of $b$

**Question 59.** It is given that $X \sim \mathrm{N}(50, 4)$ and $Y \sim \mathrm{N}(60, 9)$ where

$$\mathrm{P}(X < a) = \mathrm{P}(Y > a) = b$$

Find the value of $a$ and the value of $b$

**Question 60.** It is given that $X \sim \mathrm{N}(200, 25)$ and $Y \sim \mathrm{N}(250, 49)$ where

$$\mathrm{P}(X < a - 15) = \mathrm{P}(Y > a + 15) = b$$

Find the value of $a$ and the value of $b$

**Question 61. Continuity Corrections** The binomial distribution $X \sim \mathrm{B}(600, 0.52)$ is to be approximated using the normal distribution $Y \sim \mathrm{N}(312, 149.76)$. Approximate each of these probabilities.

a) $\mathrm{P}(X \leq 300)$
b) $\mathrm{P}(X > 320)$
c) $\mathrm{P}(X < 311)$
d) $\mathrm{P}(X \geq 295)$
e) $\mathrm{P}(X = 312)$

**Question 62.** Given that $X \sim \mathrm{B}(350, 0.6)$:

a) Find $\mathrm{P}(X \leq 200)$
b) Use a normal distribution, $Y$, to approximate your answer to a).
c) Find the percentage error between your answers to a) and b) to 3 significant figures.

**Question 63.** Given that $X \sim \mathrm{B}(540, 0.47)$:

a) Find $\mathrm{P}(X < 260)$

b) Use a normal distribution, $Y$, to approximate your answer to a).

c) Find the percentage error between your answers to a) and b) to 3 significant figures.

**Question 64.** Given that $X \sim \mathrm{B}(2000, 0.7)$:

a) Find $\mathrm{P}(X \geq 1440)$

b) Use a normal distribution, $Y$, to approximate your answer to a).

c) Find the percentage error between your answers to a) and b) to 3 significant figures.

**Question 65.** It is known that 4.5% of adults aged 18 and over use e-cigarettes.

a) In a random sample of 5000 adults, find the probability that:

i) More than 230 use e-cigarettes.

ii) More than 230 use e-cigarettes, given that less than 250 do.

b) Use a normal approximation to estimate your answers to a) i) and a) ii).